

1 Summary

Model (Run-name): model3 (test)

Iteration: 5

TAC: 62799.02782479629

Took: 53.23299598693848s

Total Time: 103.89574551582336s

Epsilons:

- ε_{LMTD} : 1e-06
- $\varepsilon_{Balancing}$: 1e-06
- ε_{Area} : 1e-06
- ε_{Beta} : 1e-06

Gurobi Tolerances:

- IntFeasTol: 1e-05
- FeasibilityTol: 1e-06
- OptimalityTol: 1e-06

Gamsworld Solution: http://www.gamsworld.org/minlp/minlplib2/html/heatexch_gen3.html

2 Results

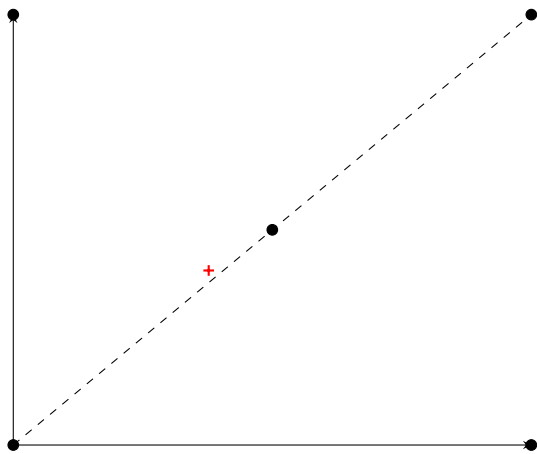
Variable	Value	Variable	Value	Variable	Value
$z_{1,3,2}$	1.0	$dt_{2,2,2}$	22.2	$RecLMTD_{2,3,1}$	0.01704103250528602
$z_{2,2,1}$	1.0	$dt_{2,3,1}$	27.8	$RecLMTD_{3,4,1}$	0.018848804502494772
$z_{2,3,1}$	1.0	$dt_{2,3,2}$	30.123809523809523	$RecLMTD_{4,2,1}$	0.02792261819314902
$z_{3,4,1}$	1.0	$dt_{3,4,1}$	50.0	$RecLMTD_{4,5,1}$	0.02478415534899077
$z_{4,2,1}$	1.0	$dt_{3,4,2}$	34.03716216216215	$RecLMTD_{5,1,1}$	0.014698147111262275
$z_{4,5,1}$	1.0	$dt_{4,2,1}$	49.4	$RecLMTD_{5,5,2}$	0.036354527697212716
$z_{5,1,1}$	1.0	$dt_{4,2,2}$	10.0	$RecLMTD_{cu,3}$	0.01764040015417513
$z_{5,5,2}$	1.0	$dt_{4,5,1}$	66.7	$RecLMTD_{cu,5}$	0.015930056104694505
$z_{cu,3}$	1.0	$dt_{4,5,2}$	10.0	$q_{1,3,2}$	586.96
$z_{cu,5}$	1.0	$dt_{5,1,1}$	38.9	$q_{2,2,1}$	224.9
$A_{1,3,2}^\beta$	2.2740842795301446	$dt_{5,1,2}$	95.96214689265534	$q_{2,3,1}$	952.76
$A_{2,2,1}^\beta$	0.6930935849085421	$dt_{5,5,2}$	31.95135552575042	$q_{3,4,1}$	1634.85
$A_{2,3,1}^\beta$	0.7386453315863175	$dt_{5,5,3}$	10.0	$q_{4,2,1}$	422.309999999999983
$A_{3,4,1}^\beta$	4.423023039000306	$dt_{cu,3}$	76.23716216216215	$q_{4,5,1}$	1117.4100000000003
$A_{4,2,1}^\beta$	0.5647178675988396	$dt_{cu,5}$	91.84463276836158	$q_{5,1,1}$	760.0
$A_{4,5,1}^\beta$	0.9239590873044407	$f_{1,3,2}^H$	8.8	$q_{5,5,2}$	426.8800000000001
$A_{5,1,1}^\beta$	3.058353873744494	$f_{2,2,1}^H$	2.024302430243015	$q_{cu,3}$	749.43
$A_{5,5,2}^\beta$	0.5936069685056475	$f_{2,3,1}^H$	8.575697569756985	$q_{cu,5}$	1172.5299999999997
$A_{cu,3}^\beta$	4.9450767456854265	$f_{3,4,1}^H$	10.14804469273743	$q_{1,1}^{(H)}$	160.0
$A_{cu,5}^\beta$	6.302068159096826	$f_{4,2,1}^H$	3.4558919803600636	$q_{1,2}^{(H)}$	160.0
$A_{1,3,2}$	12.373210120217555	$f_{4,5,1}^H$	9.144108019639935	$q_{1,3}^{(H)}$	93.3
$A_{2,2,1}$	1.9849079917037988	$f_{5,1,1}^H$	5.701425356339083	$q_{2,1}^{(H)}$	248.9
$A_{2,3,1}$	5.309777914007857	$f_{5,5,2}^H$	3.4558542241249204	$q_{2,2}^{(H)}$	137.8
$A_{3,4,1}$	36.252903577533615	$f_{1,3,2}^C$	8.4	$q_{2,3}^{(H)}$	137.8
$A_{4,2,1}$	3.1950822772839023	$f_{2,2,1}^C$	2.1196983977379844	$q_{3,1}^{(H)}$	226.7
$A_{4,5,1}$	7.393700787401576	$f_{2,3,1}^C$	8.4	$q_{3,2}^{(H)}$	116.23716216216215
$A_{5,1,1}$	13.141872711246277	$f_{3,4,1}^C$	17.3	$q_{3,3}^{(H)}$	116.23716216216215
$A_{5,5,2}$	4.755793226381462	$f_{4,2,1}^C$	3.9803016022620152	$q_{4,1}^{(H)}$	271.1
$A_{cu,3}$	14.376341772690381	$f_{4,5,1}^C$	10.05769576957696	$q_{4,2}^{(H)}$	148.9
$A_{cu,5}$	21.50633939847323	$f_{5,1,1}^C$	7.6	$q_{4,3}^{(H)}$	148.9
$dt_{1,3,2}$	52.32380952380952	$f_{5,5,2}^C$	13.9	$q_{5,1}^{(H)}$	198.9
$dt_{1,3,3}$	55.5	$RecLMTD_{1,3,2}$	0.017918135140699402	$q_{5,2}^{(H)}$	155.96214689265537
$dt_{2,2,1}$	27.2	$RecLMTD_{2,2,1}$	0.04062493688474317	$q_{5,3}^{(H)}$	131.84463276836158

Variable	Value	Variable	Value
$t_{1,1}^{(C)}$	160.0	$b_{2,3,1}^{H,\text{in}}$	2134.4911251125136
$t_{1,2}^{(C)}$	60.0	$b_{3,4,1}^{H,\text{in}}$	2300.561731843575
$t_{1,3}^{(C)}$	60.0	$b_{4,2,1}^{H,\text{in}}$	936.8923158756133
$t_{2,1}^{(C)}$	221.7	$b_{4,5,1}^{H,\text{in}}$	2478.967684124387
$t_{2,2}^{(C)}$	115.6	$b_{5,1,1}^{H,\text{in}}$	1134.0135033758438
$t_{2,3}^{(C)}$	115.6	$b_{5,5,2}^{H,\text{in}}$	653.584037102595
$t_{3,1}^{(C)}$	221.1	$b_{1,3,2}^{H,\text{out}}$	821.0400000000001
$t_{3,2}^{(C)}$	107.67619047619047	$b_{2,2,1}^{H,\text{out}}$	278.94887488748753
$t_{3,3}^{(C)}$	37.8	$b_{2,3,1}^{H,\text{out}}$	1181.7311251125136
$t_{4,1}^{(C)}$	176.7	$b_{3,4,1}^{H,\text{out}}$	665.7117318435753
$t_{4,2}^{(C)}$	82.2	$b_{4,2,1}^{H,\text{out}}$	514.5823158756135
$t_{4,3}^{(C)}$	82.2	$b_{4,5,1}^{H,\text{out}}$	1361.5576841243865
$t_{5,1}^{(C)}$	204.4	$b_{5,1,1}^{H,\text{out}}$	374.0135033758438
$t_{5,2}^{(C)}$	124.01079136690647	$b_{5,5,2}^{H,\text{out}}$	226.70403710259478
$t_{5,3}^{(C)}$	93.3	$b_{1,3,2}^{C,\text{in}}$	317.52
$t_{1,3,2}^H$	93.3	$b_{2,2,1}^{C,\text{in}}$	245.03713477851215
$t_{2,2,1}^H$	137.8	$b_{2,3,1}^{C,\text{in}}$	904.48
$t_{2,3,1}^H$	159.0169811320755	$b_{3,4,1}^{C,\text{in}}$	1422.06
$t_{3,4,1}^H$	116.23716216216218	$b_{4,2,1}^{C,\text{in}}$	460.12286522148895
$t_{4,2,1}^H$	237.58333333333334	$b_{4,5,1}^{C,\text{in}}$	938.3830153015302
$t_{4,5,1}^H$	182.41666666666666	$b_{5,1,1}^{C,\text{in}}$	456.0
$t_{5,1,1}^H$	65.6	$b_{5,5,2}^{C,\text{in}}$	1296.87
$t_{5,5,2}^H$	65.6	$b_{1,3,2}^{C,\text{out}}$	904.48
$t_{1,3,2}^C$	107.67619047619047	$b_{2,2,1}^{C,\text{out}}$	469.9371347785111
$t_{2,2,1}^C$	152.4688524590164	$b_{2,3,1}^{C,\text{out}}$	1857.24
$t_{2,3,1}^C$	221.1	$b_{3,4,1}^{C,\text{out}}$	3056.91
$t_{3,4,1}^C$	176.7	$b_{4,2,1}^{C,\text{out}}$	882.4328652214888
$t_{4,2,1}^C$	184.83114754098358	$b_{4,5,1}^{C,\text{out}}$	2055.7930153015304
$t_{4,5,1}^C$	173.68920863309353	$b_{5,1,1}^{C,\text{out}}$	1216.0
$t_{5,1,1}^C$	160.0	$b_{5,5,2}^{C,\text{out}}$	1723.75
$t_{5,5,2}^C$	124.01079136690646		
$b_{1,3,2}^{H,\text{in}}$	1408.0		
$b_{2,2,1}^{H,\text{in}}$	503.8488748874865		

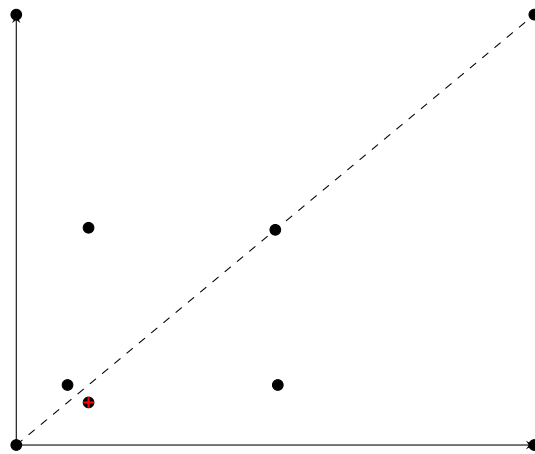
3 *RecLMTD* Tangent Points

- (1, 3, 2)
 - Value: 0.017918
 - Error: 0.000636011626145
 - Relative error: 0.034278678191874
- (2, 2, 1)
 - Value: 0.040625
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000010
- (2, 3, 1)
 - Value: 0.017041
 - Error: 0.017505625311972
 - Relative error: 0.506724135358390
- (3, 4, 1)
 - Value: 0.018849
 - Error: 0.005242781019081
 - Relative error: 0.217618762135271
- (4, 2, 1)
 - Value: 0.027923
 - Error: 0.012619649096187
 - Relative error: 0.311271419679734
- (4, 5, 1)
 - Value: 0.024784
 - Error: 0.008683567048320
 - Relative error: 0.259460950023235
- (5, 1, 1)
 - Value: 0.014698
 - Error: 0.001125995670863
 - Relative error: 0.071156819447740
- (5, 5, 2)
 - Value: 0.036355
 - Error: 0.016563822244225
 - Relative error: 0.313007156545049
- $cu, 3$
 - Value: 0.017640
 - Error: 0.000040047204437
 - Relative error: 0.002265056060237

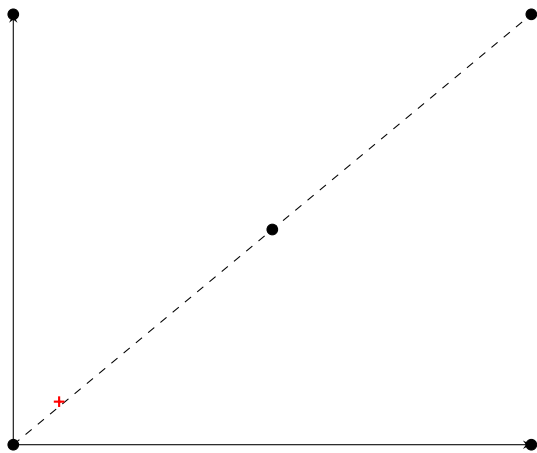
- $cu, 5$
 - Value: 0.015930
 - Error: 0.000000008460071
 - Relative error: 0.000000531075747



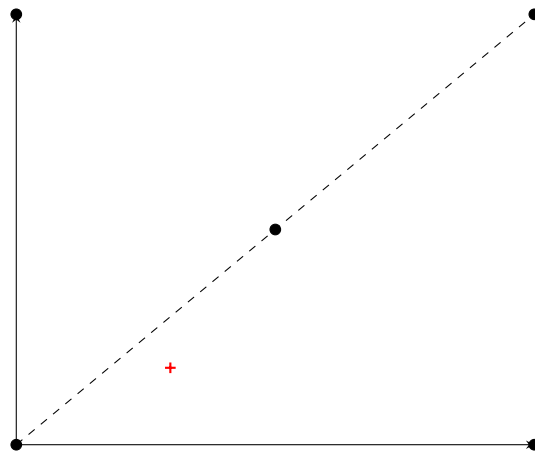
(a) *RecLMTD* - (1, 3, 2)



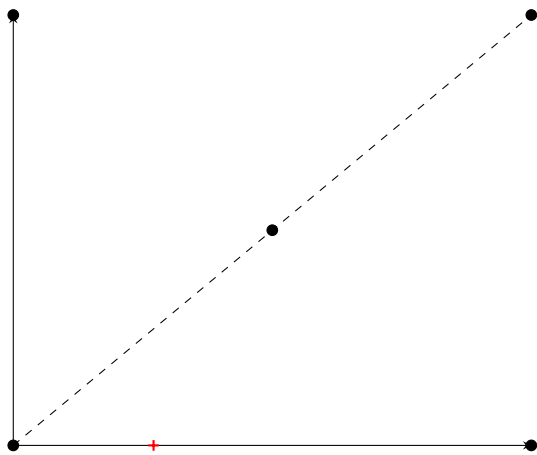
(b) *RecLMTD* - (2, 2, 1)



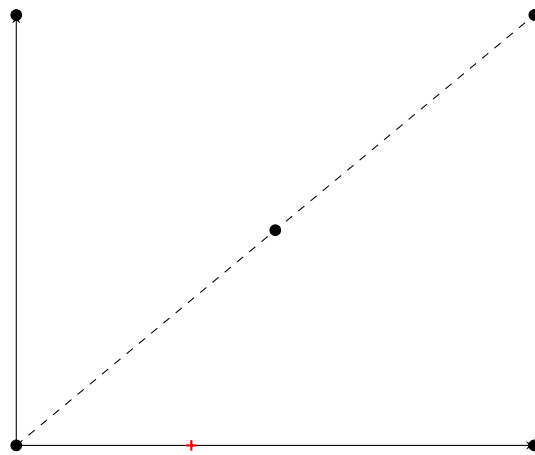
(c) *RecLMTD* - (2, 3, 1)



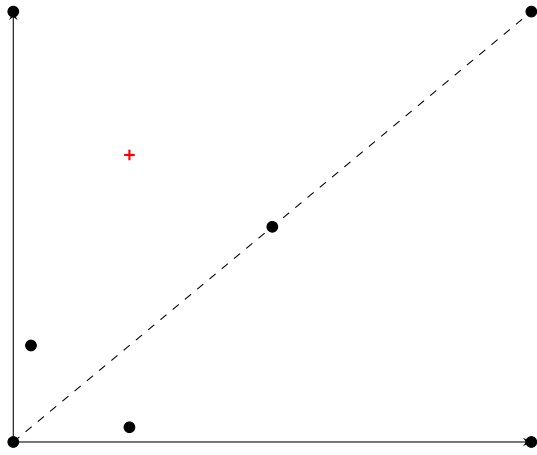
(d) *RecLMTD* - (3, 4, 1)



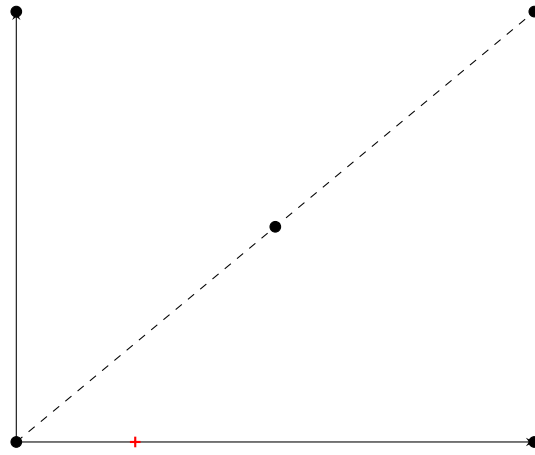
(e) *RecLMTD* - (4, 2, 1)



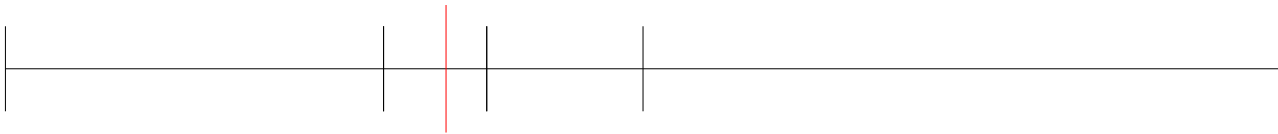
(f) *RecLMTD* - (4, 5, 1)



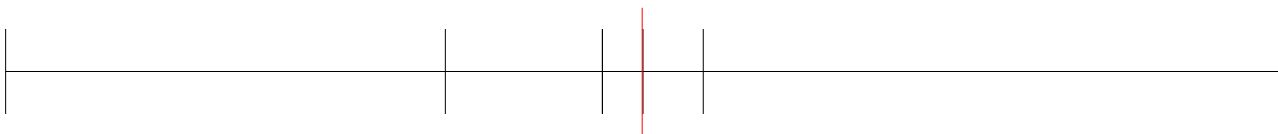
(g) *RecLMTD* - (5, 1, 1)



(h) *RecLMTD* - (5, 5, 2)



(a) *RecLMTD* - *cu*, 3



(b) *RecLMTD* - *cu*, 5

4 Balancing Breakpoints

4.1 $b^{H,\text{in}}$

- (1, 3, 2)
 - Value: 1408.0
 - Absolute error: 0.0
 - Relative error: 0.0
- (2, 2, 1)
 - Value: 503.8488748874865
 - Absolute error: 5.6843418860802e-14
 - Relative error: 1.128183899854924e-16
- (2, 3, 1)
 - Value: 2134.4911251125136
 - Absolute error: 0.0
 - Relative error: 0.0
- (3, 4, 1)
 - Value: 2300.561731843575
 - Absolute error: 0.0
 - Relative error: 0.0
- (4, 2, 1)
 - Value: 936.8923158756133
 - Absolute error: 0.0
 - Relative error: 0.0
- (4, 5, 1)
 - Value: 2478.967684124387
 - Absolute error: 0.0
 - Relative error: 0.0
- (5, 1, 1)
 - Value: 1134.0135033758438
 - Absolute error: 2.2737367544323206e-13
 - Relative error: 2.0050349909093993e-16
- (5, 5, 2)
 - Value: 653.584037102595
 - Absolute error: 114.60159296002053
 - Relative error: 0.21262583634302107

4.2 $b^{H,out}$

- (1, 3, 2)
 - Value: 821.0400000000001
 - Absolute error: 0.0
 - Relative error: 0.0
- (2, 2, 1)
 - Value: 278.94887488748753
 - Absolute error: 5.684341886080802e-14
 - Relative error: 2.0377719352241697e-16
- (2, 3, 1)
 - Value: 1181.7311251125136
 - Absolute error: 181.95041353191846
 - Relative error: -0.1334258830788208
- (3, 4, 1)
 - Value: 665.7117318435753
 - Absolute error: 513.8681847350147
 - Relative error: -0.4356366003801642
- (4, 2, 1)
 - Value: 514.5823158756135
 - Absolute error: 306.480020458265
 - Relative error: -0.3732725359522974
- (4, 5, 1)
 - Value: 1361.5576841243865
 - Absolute error: 306.480020458265
 - Relative error: -0.18373686614892637
- (5, 1, 1)
 - Value: 374.0135033758438
 - Absolute error: 0.0
 - Relative error: 0.0
- (5, 5, 2)
 - Value: 226.70403710259478
 - Absolute error: 2.842170943040401e-14
 - Relative error: 1.2536922497565309e-16

4.3 $b^{C,in}$

- (1, 3, 2)
 - Value: 317.52
 - Absolute error: 0.0
 - Relative error: 0.0
- (2, 2, 1)
 - Value: 245.03713477851215
 - Absolute error: 1.1652900866465643e-12
 - Relative error: 4.755565264423573e-15
- (2, 3, 1)
 - Value: 904.48
 - Absolute error: 0.0
 - Relative error: 0.0
- (3, 4, 1)
 - Value: 1422.06
 - Absolute error: 2.2737367544323206e-13
 - Relative error: -1.598903530394161e-16
- (4, 2, 1)
 - Value: 460.12286522148895
 - Absolute error: 0.0
 - Relative error: 0.0
- (4, 5, 1)
 - Value: 938.3830153015302
 - Absolute error: 308.87979641129584
 - Relative error: -0.24764612037708486
- (5, 1, 1)
 - Value: 456.0
 - Absolute error: 0.0
 - Relative error: 0.0
- (5, 5, 2)
 - Value: 1296.87
 - Absolute error: 0.0
 - Relative error: 0.0

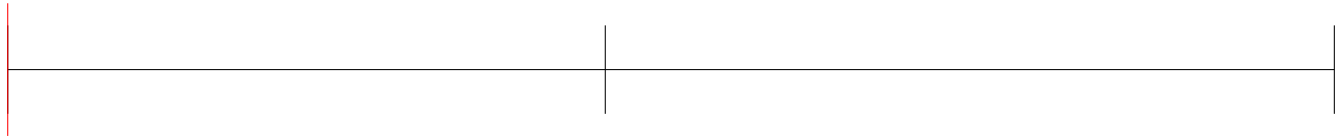
4.4 $b^{C,out}$

- (1, 3, 2)
 - Value: 904.48
 - Absolute error: 0.0
 - Relative error: 0.0
- (2, 2, 1)
 - Value: 469.9371347785111
 - Absolute error: 146.7491525161849
 - Relative error: 0.4540674795174503
- (2, 3, 1)
 - Value: 1857.24
 - Absolute error: 0.0
 - Relative error: 0.0
- (3, 4, 1)
 - Value: 3056.91
 - Absolute error: 0.0
 - Relative error: 0.0
- (4, 2, 1)
 - Value: 882.4328652214888
 - Absolute error: 146.74915251618495
 - Relative error: 0.19947315671370439
- (4, 5, 1)
 - Value: 2055.7930153015304
 - Absolute error: 308.87979641129573
 - Relative error: 0.17681461967957313
- (5, 1, 1)
 - Value: 1216.0
 - Absolute error: 0.0
 - Relative error: 0.0
- (5, 5, 2)
 - Value: 1723.75
 - Absolute error: 2.2737367544323206e-13
 - Relative error: 1.31906410699482e-16

A horizontal number line is shown. A red vertical line is drawn at the position of 0. A black vertical line is drawn at the position of 1. The number line is labeled with 0 and 1.



(a) $t^{(H)} - (1, 3, 2)$



(a) $t^{(H)} - (2, 2, 1)$



(a) $t^{(H)} - (2, 3, 1)$



(a) $t^{(H)} - (3, 4, 1)$



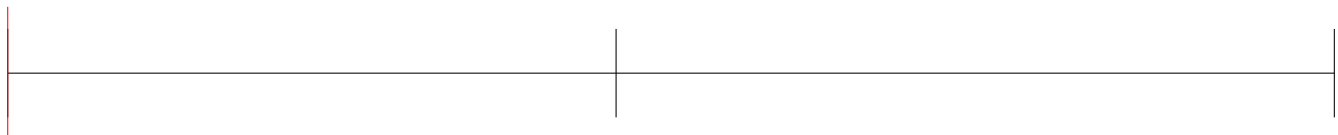
(a) $t^{(H)} - (4, 2, 1)$



(a) $t^{(H)} - (4, 5, 1)$



(a) $t^{(H)} - (5, 1, 1)$



(a) $t^{(H)} - (5, 5, 2)$



$$(a) \ t^{(C)} - (3, 3)$$



$$(a) \ t^{(C)} - (2, 2)$$



$$(a) \ t^{(C)} - (3, 2)$$



$$(a) \ t^{(C)} - (4, 2)$$



$$(a) \ t^{(C)} - (2, 2)$$



$$(a) \ t^{(C)} - (5, 2)$$



$$(a) \ t^{(C)} - (1, 2)$$



$$(a) \ t^{(C)} - (5, 3)$$



$$(a) \ t^{(C)} - (1, 3, 2)$$



$$(a) \ t^{(C)} - (2, 2, 1)$$



$$(a) \ t^{(C)} - (2, 3, 1)$$



$$(a) \ t^{(C)} - (3, 4, 1)$$



$$(a) \ t^{(C)} - (4, 2, 1)$$



$$(a) \ t^{(C)} - (4, 5, 1)$$



$$(a) \ t^{(C)} - (5, 1, 1)$$



$$(a) \ t^{(C)} - (5, 5, 2)$$

5 Area Breakpoints

5.1 A

- $(1, 3, 2)$
 - Value: 12.373210
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000000
- $(2, 2, 1)$
 - Value: 1.984908
 - Error: 8.763972367565305
 - Relative error: 0.815338163105318
- $(2, 3, 1)$
 - Value: 5.309778
 - Error: 13.791415179799564
 - Relative error: 0.722018520626899
- $(3, 4, 1)$
 - Value: 36.252904
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000000
- $(4, 2, 1)$
 - Value: 3.195082
 - Error: 10.677859945244050
 - Relative error: 0.769689642901022
- $(4, 5, 1)$
 - Value: 7.393701
 - Error: 25.187549834381699
 - Relative error: 0.773068846459249
- $(5, 1, 1)$
 - Value: 13.141873
 - Error: 0.0000000000000007
 - Relative error: -0.0000000000000001
- $(5, 5, 2)$
 - Value: 4.755793
 - Error: 13.501878283484620
 - Relative error: 0.739518085654486

5.2 A_{CU}

- $cu, 3$
 - Value: 14.376342
 - Error: 1.176887742066640
 - Relative error: 0.075668383916665
- $cu, 5$
 - Value: 21.506339
 - Error: 0.468329642041411
 - Relative error: 0.021312250081125



(a) $A^\beta - (1, 3, 2)$



(a) $A^\beta - (2, 2, 1)$



(a) $A^\beta - (2, 3, 1)$



(a) $A^\beta - (3, 4, 1)$



(a) $A^\beta - (4, 2, 1)$



(a) $A^\beta - (4, 5, 1)$



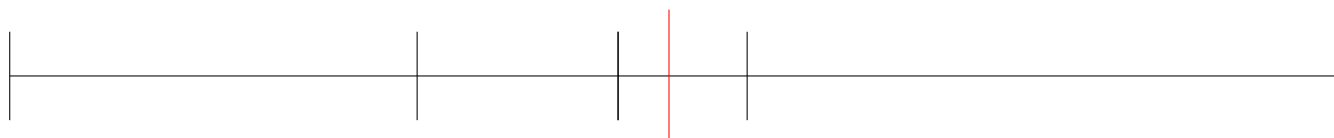
(a) $A^\beta - (5, 1, 1)$



(a) $A^\beta - (5, 5, 2)$



(a) $A^\beta - cu, 3$

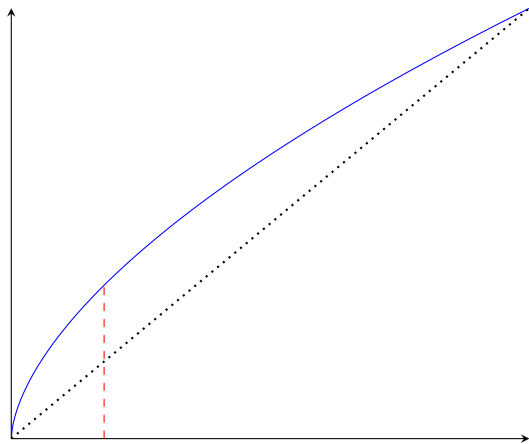


(a) $A^\beta - cu, 5$

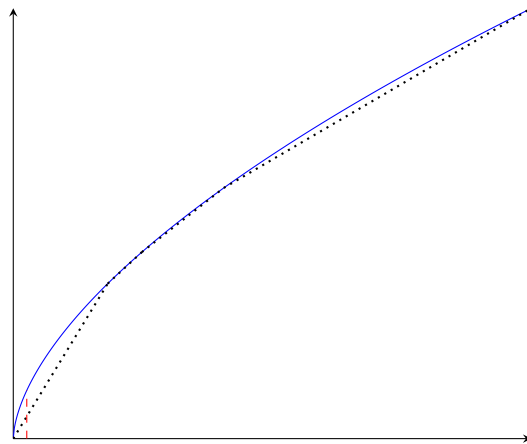
6 A^β Breakpoints

- (1, 3, 2)
 - q : 12.373210
 - $\hat{q}^\beta$: 2.274084
 - q^β : 4.523655
 - Breakpoints: [0, 69.05411764705883]
 - Error: 2.249570274198942
 - Relative error: 0.497290464486176
- (2, 2, 1)
 - q : 1.984908
 - $\hat{q}^\beta$: 0.693094
 - q^β : 1.508844
 - Breakpoints: [0, 13.879417919656465, 18.890814020083003, 30.93278282491032, 76.14235294117645]
 - Error: 0.815750025880726
 - Relative error: 0.540645842980381
- (2, 3, 1)
 - q : 5.309778
 - $\hat{q}^\beta$: 0.738645
 - q^β : 2.722989
 - Breakpoints: [0, 138.54823529411763]
 - Error: 1.984343211343141
 - Relative error: 0.728737260571921
- (3, 4, 1)
 - q : 36.252904
 - $\hat{q}^\beta$: 4.423023
 - q^β : 8.621954
 - Breakpoints: [0, 192.3352941176471]
 - Error: 4.198930486347887
 - Relative error: 0.487004537197191
- (4, 2, 1)
 - q : 3.195082
 - $\hat{q}^\beta$: 0.564718
 - q^β : 2.007656
 - Breakpoints: [0, 76.14235294117645]
 - Error: 1.442937793547967
 - Relative error: 0.718717766932073
- (4, 5, 1)

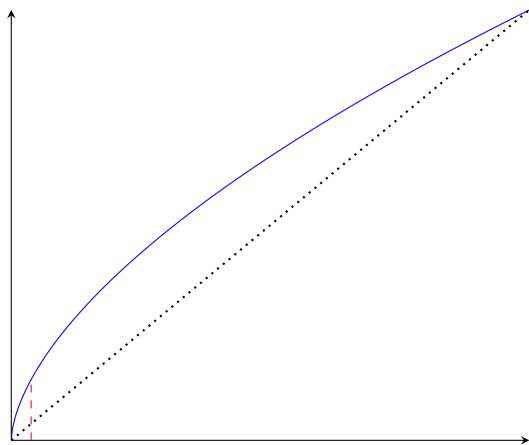
- q : 7.393701
- $\hat{q}^\beta$: 0.923959
- q^β : 3.321369
- Breakpoints: [0, 181.14352941176475]
- Error: 2.397409873890228
- Relative error: 0.721813776759057
- (5, 1, 1)
 - q : 13.141873
 - $\hat{q}^\beta$: 3.058354
 - q^β : 4.690232
 - Breakpoints: [0, 1.4656333375682882, 89.41176470588236]
 - Error: 1.631878433391785
 - Relative error: 0.347931259376822
- (5, 5, 2)
 - q : 4.755793
 - $\hat{q}^\beta$: 0.593607
 - q^β : 2.548788
 - Breakpoints: [0, 181.68117647058827]
 - Error: 1.955181410863774
 - Relative error: 0.767102293265906
- $cu, 3$
 - q : 14.376342
 - $\hat{q}^\beta$: 4.945077
 - q^β : 4.949816
 - Breakpoints: [0, 7.499856372317694, 10.320104112088831, 13.6435791010403, 16.738997089244645, 128.44338870924773]
 - Error: 0.004739005087684
 - Relative error: 0.000957410401982
- $cu, 5$
 - q : 21.506339
 - $\hat{q}^\beta$: 6.302068
 - q^β : 6.302892
 - Breakpoints: [0, 9.832103793393394, 14.682070969056184, 20.944873037188852, 22.410514202757604, 127.1036185995295]
 - Error: 0.000824299518494
 - Relative error: 0.000130781149116



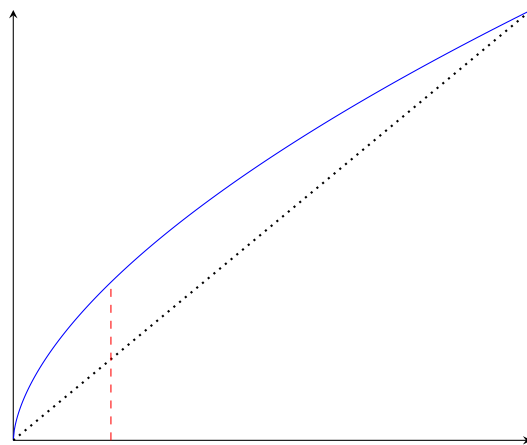
(a) $q^\beta - (1, 3, 2)$



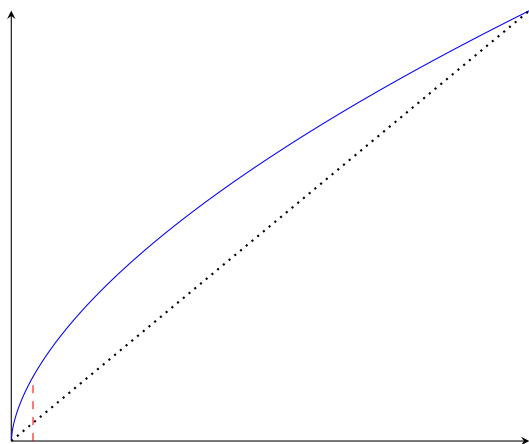
(b) $q^\beta - (2, 2, 1)$



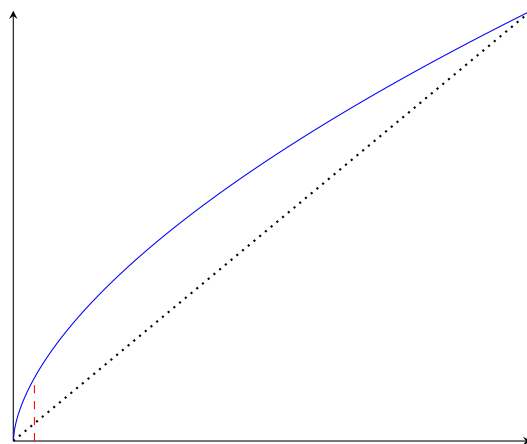
(c) $q^\beta - (2, 3, 1)$



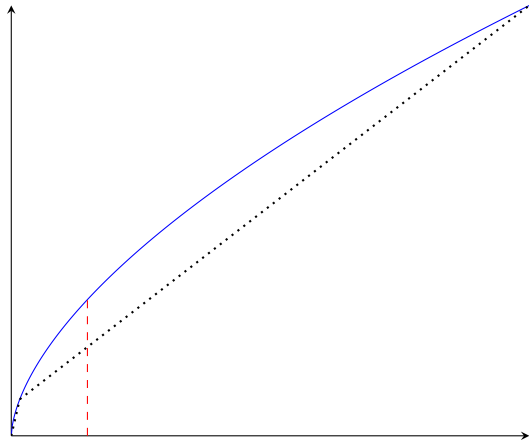
(d) $q^\beta - (3, 4, 1)$



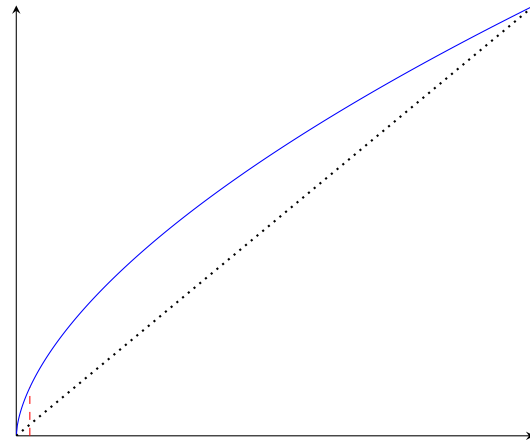
(e) $q^\beta - (4, 2, 1)$



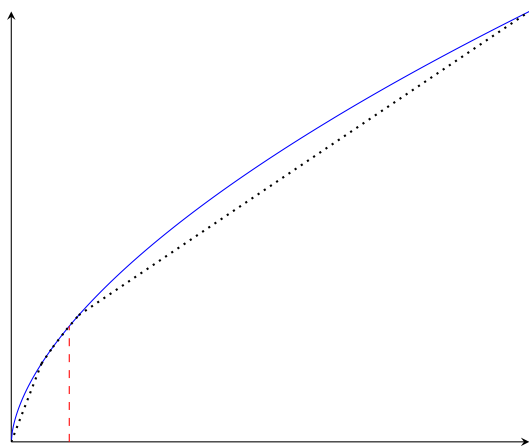
(f) $q^\beta - (4, 5, 1)$



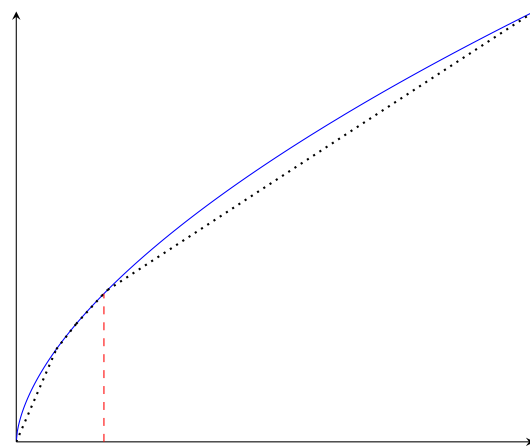
(g) $q^\beta - (5, 1, 1)$



(h) $q^\beta - (5, 5, 2)$



(a) $q^\beta - cu, 3$



(b) $q^\beta - cu, 5$

7 Other variables

Variable	Value	Variable	Value	Variable	Value
$z_{1,1,1}$	0.0	$z_{4,5,2}$	-0.0	$A_{2,5,2}^\beta$	0.0
$z_{1,1,2}$	0.0	$z_{5,1,2}$	-0.0	$A_{3,1,1}^\beta$	0.0
$z_{1,2,1}$	0.0	$z_{5,2,1}$	0.0	$A_{3,1,2}^\beta$	0.0
$z_{1,2,2}$	-0.0	$z_{5,2,2}$	-0.0	$A_{3,2,1}^\beta$	0.0
$z_{1,3,1}$	0.0	$z_{5,3,1}$	0.0	$A_{3,2,2}^\beta$	0.0
$z_{1,4,1}$	0.0	$z_{5,3,2}$	-0.0	$A_{3,3,1}^\beta$	0.0
$z_{1,4,2}$	-0.0	$z_{5,4,1}$	-0.0	$A_{3,3,2}^\beta$	0.0
$z_{1,5,1}$	0.0	$z_{5,4,2}$	0.0	$A_{3,4,2}^\beta$	0.0
$z_{1,5,2}$	-0.0	$z_{5,5,1}$	0.0	$A_{3,5,1}^\beta$	0.0
$z_{2,1,1}$	-0.0	$z_{cu,1}$	-0.0	$A_{3,5,2}^\beta$	0.0
$z_{2,1,2}$	-0.0	$z_{cu,2}$	-0.0	$A_{4,1,1}^\beta$	0.0
$z_{2,2,2}$	0.0	$z_{cu,4}$	-0.0	$A_{4,1,2}^\beta$	0.0
$z_{2,3,2}$	-0.0	$z_{hu,1}$	-0.0	$A_{4,2,2}^\beta$	0.0
$z_{2,4,1}$	-0.0	$z_{hu,2}$	-0.0	$A_{4,3,1}^\beta$	0.0
$z_{2,4,2}$	-0.0	$z_{hu,3}$	-0.0	$A_{4,3,2}^\beta$	0.0
$z_{2,5,1}$	-0.0	$z_{hu,4}$	-0.0	$A_{4,4,1}^\beta$	0.0
$z_{2,5,2}$	-0.0	$z_{hu,5}$	-0.0	$A_{4,4,2}^\beta$	0.0
$z_{3,1,1}$	-0.0	$A_{1,1,1}^\beta$	0.0	$A_{4,5,2}^\beta$	0.0
$z_{3,1,2}$	-0.0	$A_{1,1,2}^\beta$	0.0	$A_{5,1,2}^\beta$	0.0
$z_{3,2,1}$	-0.0	$A_{1,2,1}^\beta$	0.0	$A_{5,2,1}^\beta$	0.0
$z_{3,2,2}$	-0.0	$A_{1,2,2}^\beta$	0.0	$A_{5,2,2}^\beta$	0.0
$z_{3,3,1}$	-0.0	$A_{1,3,1}^\beta$	0.0	$A_{5,3,1}^\beta$	0.0
$z_{3,3,2}$	-0.0	$A_{1,4,1}^\beta$	0.0	$A_{5,3,2}^\beta$	0.0
$z_{3,4,2}$	-0.0	$A_{1,4,2}^\beta$	0.0	$A_{5,4,1}^\beta$	0.0
$z_{3,5,1}$	-0.0	$A_{1,5,1}^\beta$	0.0	$A_{5,4,2}^\beta$	0.0
$z_{3,5,2}$	0.0	$A_{1,5,2}^\beta$	0.0	$A_{5,5,1}^\beta$	0.0
$z_{4,1,1}$	-0.0	$A_{2,1,1}^\beta$	0.0	$A_{cu,1}^\beta$	0.0
$z_{4,1,2}$	-0.0	$A_{2,1,2}^\beta$	0.0	$A_{cu,2}^\beta$	0.0
$z_{4,2,2}$	-0.0	$A_{2,2,2}^\beta$	0.0	$A_{cu,4}^\beta$	0.0
$z_{4,3,1}$	-0.0	$A_{2,3,2}^\beta$	0.0	$A_{hu,1}^\beta$	0.0
$z_{4,3,2}$	-0.0	$A_{2,4,1}^\beta$	0.0	$A_{hu,2}^\beta$	0.0
$z_{4,4,1}$	0.0	$A_{2,4,2}^\beta$	0.0	$A_{hu,3}^\beta$	0.0
$z_{4,4,2}$	-0.0	$A_{2,5,1}^\beta$	0.0	$A_{hu,4}^\beta$	0.0

Variable	Value	Variable	Value	Variable	Value
$A_{hu,5}^\beta$	0.0	$A_{4,4,2}$	0.0	$dt_{2,1,3}$	100.0
$A_{1,1,1}$	0.0	$A_{4,5,2}$	0.0	$dt_{2,2,3}$	106.1
$A_{1,1,2}$	0.0	$A_{5,1,2}$	0.0	$dt_{2,3,3}$	183.3
$A_{1,2,1}$	0.0	$A_{5,2,1}$	0.0	$dt_{2,4,1}$	111.1
$A_{1,2,2}$	0.0	$A_{5,2,2}$	0.0	$dt_{2,4,2}$	10.0
$A_{1,3,1}$	0.0	$A_{5,3,1}$	0.0	$dt_{2,4,3}$	94.5
$A_{1,4,1}$	0.0	$A_{5,3,2}$	0.0	$dt_{2,5,1}$	111.1
$A_{1,4,2}$	0.0	$A_{5,4,1}$	0.0	$dt_{2,5,2}$	10.0
$A_{1,5,1}$	0.0	$A_{5,4,2}$	0.0	$dt_{2,5,3}$	111.1
$A_{1,5,2}$	0.0	$A_{5,5,1}$	0.0	$dt_{3,1,1}$	161.1
$A_{2,1,1}$	0.0	$A_{cu,1}$	0.0	$dt_{3,1,2}$	150.63716216216216
$A_{2,1,2}$	0.0	$A_{cu,2}$	0.0	$dt_{3,1,3}$	10.0
$A_{2,2,2}$	0.0	$A_{cu,4}$	0.0	$dt_{3,2,1}$	111.1
$A_{2,3,2}$	0.0	$A_{hu,1}$	0.0	$dt_{3,2,2}$	10.0
$A_{2,4,1}$	0.0	$A_{hu,2}$	0.0	$dt_{3,2,3}$	10.0
$A_{2,4,2}$	0.0	$A_{hu,3}$	0.0	$dt_{3,3,1}$	161.1
$A_{2,5,1}$	0.0	$A_{hu,4}$	0.0	$dt_{3,3,2}$	10.0
$A_{2,5,2}$	0.0	$A_{hu,5}$	0.0	$dt_{3,3,3}$	27.8
$A_{3,1,1}$	0.0	$dt_{1,1,1}$	66.7	$dt_{3,4,3}$	10.0
$A_{3,1,2}$	0.0	$dt_{1,1,2}$	100.0	$dt_{3,5,1}$	133.4
$A_{3,2,1}$	0.0	$dt_{1,1,3}$	100.0	$dt_{3,5,2}$	10.0
$A_{3,2,2}$	0.0	$dt_{1,2,1}$	17.15006133296988	$dt_{3,5,3}$	10.0
$A_{3,3,1}$	0.0	$dt_{1,2,2}$	17.15006133296988	$dt_{4,1,1}$	122.2
$A_{3,3,2}$	0.0	$dt_{1,2,3}$	17.15006133296988	$dt_{4,1,2}$	10.0
$A_{3,4,2}$	0.0	$dt_{1,3,1}$	66.7	$dt_{4,1,3}$	100.0
$A_{3,5,1}$	0.0	$dt_{1,4,1}$	66.7	$dt_{4,2,3}$	106.1
$A_{3,5,2}$	0.0	$dt_{1,4,2}$	10.0	$dt_{4,3,1}$	122.2
$A_{4,1,1}$	0.0	$dt_{1,4,3}$	77.8	$dt_{4,3,2}$	10.0
$A_{4,1,2}$	0.0	$dt_{1,5,1}$	66.7	$dt_{4,3,3}$	183.3
$A_{4,2,2}$	0.0	$dt_{1,5,2}$	66.7	$dt_{4,4,1}$	122.2
$A_{4,3,1}$	0.0	$dt_{1,5,3}$	66.7	$dt_{4,4,2}$	10.0
$A_{4,3,2}$	0.0	$dt_{2,1,1}$	111.1	$dt_{4,4,3}$	94.5
$A_{4,4,1}$	0.0	$dt_{2,1,2}$	10.0	$dt_{4,5,3}$	111.1

Variable	Value	Variable	Value	Variable	Value
$dt_{5,1,3}$	10.0	$f_{2,4,2}^H$	0.0	$f_{1,4,1}^C$	0.0
$dt_{5,2,1}$	74.66004781348396	$f_{2,5,1}^H$	0.0	$f_{1,4,2}^C$	0.0
$dt_{5,2,2}$	74.66004781348396	$f_{2,5,2}^H$	0.0	$f_{1,5,1}^C$	0.0
$dt_{5,2,3}$	74.66004781348391	$f_{3,1,1}^H$	4.651955307262569	$f_{1,5,2}^C$	0.0
$dt_{5,3,1}$	133.3	$f_{3,1,2}^H$	14.8	$f_{2,1,1}^C$	0.0
$dt_{5,3,2}$	53.21347072531854	$f_{3,2,1}^H$	0.0	$f_{2,1,2}^C$	0.0
$dt_{5,3,3}$	161.1	$f_{3,2,2}^H$	0.0	$f_{2,2,2}^C$	6.1
$dt_{5,4,1}$	116.7	$f_{3,3,1}^H$	0.0	$f_{2,3,2}^C$	0.0
$dt_{5,4,2}$	116.7	$f_{3,3,2}^H$	0.0	$f_{2,4,1}^C$	0.0
$dt_{5,4,3}$	116.7	$f_{3,4,2}^H$	0.0	$f_{2,4,2}^C$	0.0
$dt_{5,5,1}$	105.6	$f_{3,5,1}^H$	0.0	$f_{2,5,1}^C$	3.842304230423041
$dt_{cu,1}$	53.3	$f_{3,5,2}^H$	0.0	$f_{2,5,2}^C$	0.0
$dt_{cu,2}$	97.8	$f_{4,1,1}^H$	0.0	$f_{3,1,1}^C$	0.0
$dt_{cu,4}$	108.9	$f_{4,1,2}^H$	0.0	$f_{3,1,2}^C$	0.0
$dt_{hu,1}$	80.0	$f_{4,2,2}^H$	0.0	$f_{3,2,1}^C$	0.0
$dt_{hu,2}$	18.3	$f_{4,3,1}^H$	0.0	$f_{3,2,2}^C$	0.0
$dt_{hu,3}$	18.9	$f_{4,3,2}^H$	12.6	$f_{3,3,1}^C$	0.0
$dt_{hu,4}$	63.3	$f_{4,4,1}^H$	0.0	$f_{3,3,2}^C$	0.0
$dt_{hu,5}$	35.6	$f_{4,4,2}^H$	0.0	$f_{3,4,2}^C$	17.3
$f_{1,1,1}^H$	0.0	$f_{4,5,2}^H$	0.0	$f_{3,5,1}^C$	0.0
$f_{1,1,2}^H$	0.0	$f_{5,1,2}^H$	0.0	$f_{3,5,2}^C$	0.0
$f_{1,2,1}^H$	0.0	$f_{5,2,1}^H$	11.998574643660918	$f_{4,1,1}^C$	0.0
$f_{1,2,2}^H$	0.0	$f_{5,2,2}^H$	0.0	$f_{4,1,2}^C$	7.6
$f_{1,3,1}^H$	0.0	$f_{5,3,1}^H$	0.0	$f_{4,2,2}^C$	0.0
$f_{1,4,1}^H$	8.8	$f_{5,3,2}^H$	0.0	$f_{4,3,1}^C$	0.0
$f_{1,4,2}^H$	0.0	$f_{5,4,1}^H$	0.0	$f_{4,3,2}^C$	0.0
$f_{1,5,1}^H$	0.0	$f_{5,4,2}^H$	14.24414577587508	$f_{4,4,1}^C$	0.0
$f_{1,5,2}^H$	0.0	$f_{5,5,1}^H$	0.0	$f_{4,4,2}^C$	0.0
$f_{2,1,1}^H$	0.0	$f_{1,1,1}^C$	0.0	$f_{4,5,2}^C$	0.0
$f_{2,1,2}^H$	0.0	$f_{1,1,2}^C$	0.0	$f_{5,1,2}^C$	0.0
$f_{2,2,2}^H$	10.6	$f_{1,2,1}^C$	0.0	$f_{5,2,1}^C$	0.0
$f_{2,3,2}^H$	0.0	$f_{1,2,2}^C$	0.0	$f_{5,2,2}^C$	0.0
$f_{2,4,1}^H$	0.0	$f_{1,3,1}^C$	0.0	$f_{5,3,1}^C$	0.0

Variable	Value	Variable	Value	Variable	Value
$f_{5,3,2}^C$	0.0	$\text{RecLMTD}_{4,3,1}$	0.1	$q_{2,3,2}$	0.0
$f_{5,4,1}^C$	0.0	$\text{RecLMTD}_{4,3,2}$	0.016304051546590172	$q_{2,4,1}$	0.0
$f_{5,4,2}^C$	0.0	$\text{RecLMTD}_{4,4,1}$	0.020576623473060753	$q_{2,4,2}$	0.0
$f_{5,5,1}^C$	0.0	$\text{RecLMTD}_{4,4,2}$	0.022300296262842634	$q_{2,5,1}$	0.0
$\text{RecLMTD}_{1,1,1}$	0.1	$\text{RecLMTD}_{4,5,2}$	0.02554670259920889	$q_{2,5,2}$	0.0
$\text{RecLMTD}_{1,1,2}$	0.1	$\text{RecLMTD}_{5,1,2}$	0.1	$q_{3,1,1}$	0.0
$\text{RecLMTD}_{1,2,1}$	0.1	$\text{RecLMTD}_{5,2,1}$	0.1	$q_{3,1,2}$	0.0
$\text{RecLMTD}_{1,2,2}$	0.1	$\text{RecLMTD}_{5,2,2}$	0.1	$q_{3,2,1}$	0.0
$\text{RecLMTD}_{1,3,1}$	0.1	$\text{RecLMTD}_{5,3,1}$	0.1	$q_{3,2,2}$	0.0
$\text{RecLMTD}_{1,4,1}$	0.1	$\text{RecLMTD}_{5,3,2}$	0.028819701078955186	$q_{3,3,1}$	0.0
$\text{RecLMTD}_{1,4,2}$	0.1	$\text{RecLMTD}_{5,4,1}$	0.1	$q_{3,3,2}$	0.0
$\text{RecLMTD}_{1,5,1}$	0.1	$\text{RecLMTD}_{5,4,2}$	0.1	$q_{3,4,2}$	0.0
$\text{RecLMTD}_{1,5,2}$	0.1	$\text{RecLMTD}_{5,5,1}$	0.1	$q_{3,5,1}$	0.0
$\text{RecLMTD}_{2,1,1}$	0.1	$\text{RecLMTD}_{cu,1}$	0.01610890859262668	$q_{3,5,2}$	0.0
$\text{RecLMTD}_{2,1,2}$	0.023339478053072158	$\text{RecLMTD}_{cu,2}$	0.009423141115769086	$q_{4,1,1}$	0.0
$\text{RecLMTD}_{2,2,2}$	0.016168655950305547	$\text{RecLMTD}_{cu,4}$	0.008535688158765427	$q_{4,1,2}$	0.0
$\text{RecLMTD}_{2,3,2}$	0.009359730293730571	$\text{RecLMTD}_{hu,1}$	0.012387051800747543	$q_{4,2,2}$	0.0
$\text{RecLMTD}_{2,4,1}$	0.025165980106347276	$\text{RecLMTD}_{hu,2}$	0.045618305265862594	$q_{4,3,1}$	0.0
$\text{RecLMTD}_{2,4,2}$	0.1	$\text{RecLMTD}_{hu,3}$	0.04305097282506776	$q_{4,3,2}$	0.0
$\text{RecLMTD}_{2,5,1}$	0.022648232592241378	$\text{RecLMTD}_{hu,4}$	0.015434474530129737	$q_{4,4,1}$	0.0
$\text{RecLMTD}_{2,5,2}$	0.1	$\text{RecLMTD}_{hu,5}$	0.024141568940966463	$q_{4,4,2}$	0.0
$\text{RecLMTD}_{3,1,1}$	0.006388575395379913	$q_{1,1,1}$	0.0	$q_{4,5,2}$	0.0
$\text{RecLMTD}_{3,1,2}$	0.1	$q_{1,1,2}$	0.0	$q_{5,1,2}$	0.0
$\text{RecLMTD}_{3,2,1}$	0.1	$q_{1,2,1}$	0.0	$q_{5,2,1}$	0.0
$\text{RecLMTD}_{3,2,2}$	0.1	$q_{1,2,2}$	0.0	$q_{5,2,2}$	0.0
$\text{RecLMTD}_{3,3,1}$	0.1	$q_{1,3,1}$	0.0	$q_{5,3,1}$	0.0
$\text{RecLMTD}_{3,3,2}$	0.026450800778310225	$q_{1,4,1}$	0.0	$q_{5,3,2}$	0.0
$\text{RecLMTD}_{3,4,2}$	0.1	$q_{1,4,2}$	0.0	$q_{5,4,1}$	0.0
$\text{RecLMTD}_{3,5,1}$	0.1	$q_{1,5,1}$	0.0	$q_{5,4,2}$	0.0
$\text{RecLMTD}_{3,5,2}$	0.1	$q_{1,5,2}$	0.0	$q_{5,5,1}$	0.0
$\text{RecLMTD}_{4,1,1}$	0.020350921041481694	$q_{2,1,1}$	0.0	$q_{cu,1}$	0.0
$\text{RecLMTD}_{4,1,2}$	0.02150212335787724	$q_{2,1,2}$	0.0	$q_{cu,2}$	0.0
$\text{RecLMTD}_{4,2,2}$	0.029297885186065308	$q_{2,2,2}$	0.0	$q_{cu,4}$	0.0

Variable	Value	Variable	Value	Variable	Value
$q_{hu,1}$	0.0	$t_{4,2,2}^H$	148.9	$t_{3,2,1}^C$	115.6
$q_{hu,2}$	0.0	$t_{4,3,1}^H$	271.1	$t_{3,2,2}^C$	115.6
$q_{hu,3}$	0.0	$t_{4,3,2}^H$	148.9	$t_{3,3,1}^C$	111.5
$q_{hu,4}$	0.0	$t_{4,4,1}^H$	271.1	$t_{3,3,2}^C$	37.8
$q_{hu,5}$	0.0	$t_{4,4,2}^H$	148.9	$t_{3,4,2}^C$	82.2
$t_{1,1,1}^H$	160.0	$t_{4,5,2}^H$	148.9	$t_{3,5,1}^C$	166.23812949640245
$t_{1,1,2}^H$	93.3	$t_{5,1,2}^H$	65.6	$t_{3,5,2}^C$	93.3
$t_{1,2,1}^H$	160.0	$t_{5,2,1}^H$	198.9	$t_{4,1,1}^C$	60.0
$t_{1,2,2}^H$	160.0	$t_{5,2,2}^H$	65.6	$t_{4,1,2}^C$	60.0
$t_{1,3,1}^H$	160.0	$t_{5,3,1}^H$	198.9	$t_{4,2,2}^C$	115.6
$t_{1,4,1}^H$	160.0	$t_{5,3,2}^H$	65.6	$t_{4,3,1}^C$	37.8
$t_{1,4,2}^H$	119.38522727272726	$t_{5,4,1}^H$	198.9	$t_{4,3,2}^C$	37.8
$t_{1,5,1}^H$	160.0	$t_{5,4,2}^H$	131.84463276836158	$t_{4,4,1}^C$	82.2
$t_{1,5,2}^H$	160.0	$t_{5,5,1}^H$	198.9	$t_{4,4,2}^C$	82.2
$t_{2,1,1}^H$	198.8575471698114	$t_{1,1,1}^C$	90.20394736842101	$t_{4,5,2}^C$	93.3
$t_{2,1,2}^H$	137.8	$t_{1,1,2}^C$	144.10854085872572	$t_{5,1,2}^C$	60.0
$t_{2,2,2}^H$	137.8	$t_{1,2,1}^C$	115.6	$t_{5,2,1}^C$	115.6
$t_{2,3,2}^H$	137.8	$t_{1,2,2}^C$	115.6	$t_{5,2,2}^C$	115.6
$t_{2,4,1}^H$	248.9	$t_{1,3,1}^C$	111.5	$t_{5,3,1}^C$	111.5
$t_{2,4,2}^H$	137.8	$t_{1,4,1}^C$	102.85953757225435	$t_{5,3,2}^C$	37.8
$t_{2,5,1}^H$	198.85754716981077	$t_{1,4,2}^C$	102.85953757225434	$t_{5,4,1}^C$	172.98757761639607
$t_{2,5,2}^H$	137.8	$t_{1,5,1}^C$	148.85	$t_{5,4,2}^C$	146.03815028901732
$t_{3,1,1}^H$	226.7	$t_{1,5,2}^C$	93.3	$t_{5,5,1}^C$	148.85
$t_{3,1,2}^H$	116.23716216216215	$t_{2,1,1}^C$	129.7960526315789	$b_{1,1,1}^{H,in}$	0.0
$t_{3,2,1}^H$	226.7	$t_{2,1,2}^C$	60.0	$b_{1,1,2}^{H,in}$	0.0
$t_{3,2,2}^H$	65.6	$t_{2,2,2}^C$	115.6	$b_{1,2,1}^{H,in}$	0.0
$t_{3,3,1}^H$	226.7	$t_{2,3,2}^C$	37.8	$b_{1,2,2}^{H,in}$	0.0
$t_{3,3,2}^H$	65.6	$t_{2,4,1}^C$	117.2653566715965	$b_{1,3,1}^{H,in}$	0.0
$t_{3,4,2}^H$	65.6	$t_{2,4,2}^C$	112.86184971098264	$b_{1,4,1}^{H,in}$	1408.0
$t_{3,5,1}^H$	122.35608108108127	$t_{2,5,1}^C$	157.23125572159418	$b_{1,4,2}^{H,in}$	0.0
$t_{3,5,2}^H$	65.6	$t_{2,5,2}^C$	93.3	$b_{1,5,1}^{H,in}$	0.0
$t_{4,1,1}^H$	271.1	$t_{3,1,1}^C$	127.37864077669906	$b_{1,5,2}^{H,in}$	0.0
$t_{4,1,2}^H$	148.9	$t_{3,1,2}^C$	60.0	$b_{2,1,1}^{H,in}$	0.0

Variable	Value	Variable	Value	Variable	Value
$b_{2,1,2}^{H,in}$	0.0	$b_{1,1,2}^{H,out}$	0.0	$b_{5,1,2}^{H,out}$	0.0
$b_{2,2,2}^{H,in}$	1460.6800000000012	$b_{1,2,1}^{H,out}$	0.0	$b_{5,2,1}^{H,out}$	2386.516496624157
$b_{2,3,2}^{H,in}$	0.0	$b_{1,2,2}^{H,out}$	0.0	$b_{5,2,2}^{H,out}$	0.0
$b_{2,4,1}^{H,in}$	0.0	$b_{1,3,1}^{H,out}$	0.0	$b_{5,3,1}^{H,out}$	0.0
$b_{2,4,2}^{H,in}$	0.0	$b_{1,4,1}^{H,out}$	1408.0	$b_{5,3,2}^{H,out}$	0.0
$b_{2,5,1}^{H,in}$	0.0	$b_{1,4,2}^{H,out}$	0.0	$b_{5,4,1}^{H,out}$	0.0
$b_{2,5,2}^{H,in}$	0.0	$b_{1,5,1}^{H,out}$	0.0	$b_{5,4,2}^{H,out}$	2106.945962897405
$b_{3,1,1}^{H,in}$	1054.5982681564244	$b_{1,5,2}^{H,out}$	0.0	$b_{5,5,1}^{H,out}$	0.0
$b_{3,1,2}^{H,in}$	1720.31	$b_{2,1,1}^{H,out}$	0.0	$b_{1,1,1}^{C,in}$	0.0
$b_{3,2,1}^{H,in}$	0.0	$b_{2,1,2}^{H,out}$	0.0	$b_{1,1,2}^{C,in}$	0.0
$b_{3,2,2}^{H,in}$	0.0	$b_{2,2,2}^{H,out}$	1460.68	$b_{1,2,1}^{C,in}$	0.0
$b_{3,3,1}^{H,in}$	0.0	$b_{2,3,2}^{H,out}$	0.0	$b_{1,2,2}^{C,in}$	0.0
$b_{3,3,2}^{H,in}$	0.0	$b_{2,4,1}^{H,out}$	0.0	$b_{1,3,1}^{C,in}$	0.0
$b_{3,4,2}^{H,in}$	0.0	$b_{2,4,2}^{H,out}$	0.0	$b_{1,4,1}^{C,in}$	0.0
$b_{3,5,1}^{H,in}$	0.0	$b_{2,5,1}^{H,out}$	0.0	$b_{1,4,2}^{C,in}$	0.0
$b_{3,5,2}^{H,in}$	0.0	$b_{2,5,2}^{H,out}$	0.0	$b_{1,5,1}^{C,in}$	0.0
$b_{4,1,1}^{H,in}$	0.0	$b_{3,1,1}^{H,out}$	1054.5982681564244	$b_{1,5,2}^{C,in}$	0.0
$b_{4,1,2}^{H,in}$	0.0	$b_{3,1,2}^{H,out}$	1720.31	$b_{2,1,1}^{C,in}$	0.0
$b_{4,2,2}^{H,in}$	0.0	$b_{3,2,1}^{H,out}$	0.0	$b_{2,1,2}^{C,in}$	0.0
$b_{4,3,1}^{H,in}$	0.0	$b_{3,2,2}^{H,out}$	0.0	$b_{2,2,2}^{C,in}$	705.16
$b_{4,3,2}^{H,in}$	1876.14	$b_{3,3,1}^{H,out}$	0.0	$b_{2,3,2}^{C,in}$	0.0
$b_{4,4,1}^{H,in}$	0.0	$b_{3,3,2}^{H,out}$	0.0	$b_{2,4,1}^{C,in}$	0.0
$b_{4,4,2}^{H,in}$	0.0	$b_{3,4,2}^{H,out}$	0.0	$b_{2,4,2}^{C,in}$	0.0
$b_{4,5,2}^{H,in}$	0.0	$b_{3,5,1}^{H,out}$	0.0	$b_{2,5,1}^{C,in}$	785.3669846984699
$b_{5,1,2}^{H,in}$	0.0	$b_{3,5,2}^{H,out}$	0.0	$b_{2,5,2}^{C,in}$	0.0
$b_{5,2,1}^{H,in}$	2386.516496624157	$b_{4,1,1}^{H,out}$	0.0	$b_{3,1,1}^{C,in}$	0.0
$b_{5,2,2}^{H,in}$	0.0	$b_{4,1,2}^{H,out}$	0.0	$b_{3,1,2}^{C,in}$	0.0
$b_{5,3,1}^{H,in}$	0.0	$b_{4,2,2}^{H,out}$	0.0	$b_{3,2,1}^{C,in}$	0.0
$b_{5,3,2}^{H,in}$	0.0	$b_{4,3,1}^{H,out}$	0.0	$b_{3,2,2}^{C,in}$	0.0
$b_{5,4,1}^{H,in}$	0.0	$b_{4,3,2}^{H,out}$	1876.14	$b_{3,3,1}^{C,in}$	0.0
$b_{5,4,2}^{H,in}$	2106.9459628974046	$b_{4,4,1}^{H,out}$	0.0	$b_{3,3,2}^{C,in}$	0.0
$b_{5,5,1}^{H,in}$	0.0	$b_{4,4,2}^{H,out}$	0.0	$b_{3,4,2}^{C,in}$	1422.0600000000002
$b_{1,1,1}^{H,out}$	0.0	$b_{4,5,2}^{H,out}$	0.0	$b_{3,5,1}^{C,in}$	0.0

Variable	Value	Variable	Value
$b_{3,5,2}^{C,in}$	0.0	$b_{2,5,2}^{C,out}$	0.0
$b_{4,1,1}^{C,in}$	0.0	$b_{3,1,1}^{C,out}$	0.0
$b_{4,1,2}^{C,in}$	456.0	$b_{3,1,2}^{C,out}$	0.0
$b_{4,2,2}^{C,in}$	0.0	$b_{3,2,1}^{C,out}$	0.0
$b_{4,3,1}^{C,in}$	0.0	$b_{3,2,2}^{C,out}$	0.0
$b_{4,3,2}^{C,in}$	0.0	$b_{3,3,1}^{C,out}$	0.0
$b_{4,4,1}^{C,in}$	0.0	$b_{3,3,2}^{C,out}$	0.0
$b_{4,4,2}^{C,in}$	0.0	$b_{3,4,2}^{C,out}$	1422.0600000000002
$b_{4,5,2}^{C,in}$	0.0	$b_{3,5,1}^{C,out}$	0.0
$b_{5,1,2}^{C,in}$	0.0	$b_{3,5,2}^{C,out}$	0.0
$b_{5,2,1}^{C,in}$	0.0	$b_{4,1,1}^{C,out}$	0.0
$b_{5,2,2}^{C,in}$	0.0	$b_{4,1,2}^{C,out}$	456.0
$b_{5,3,1}^{C,in}$	0.0	$b_{4,2,2}^{C,out}$	0.0
$b_{5,3,2}^{C,in}$	0.0	$b_{4,3,1}^{C,out}$	0.0
$b_{5,4,1}^{C,in}$	0.0	$b_{4,3,2}^{C,out}$	0.0
$b_{5,4,2}^{C,in}$	0.0	$b_{4,4,1}^{C,out}$	0.0
$b_{5,5,1}^{C,in}$	0.0	$b_{4,4,2}^{C,out}$	0.0
$b_{1,1,1}^{C,out}$	0.0	$b_{4,5,2}^{C,out}$	0.0
$b_{1,1,2}^{C,out}$	0.0	$b_{5,1,2}^{C,out}$	0.0
$b_{1,2,1}^{C,out}$	0.0	$b_{5,2,1}^{C,out}$	0.0
$b_{1,2,2}^{C,out}$	0.0	$b_{5,2,2}^{C,out}$	0.0
$b_{1,3,1}^{C,out}$	0.0	$b_{5,3,1}^{C,out}$	0.0
$b_{1,4,1}^{C,out}$	0.0	$b_{5,3,2}^{C,out}$	0.0
$b_{1,4,2}^{C,out}$	0.0	$b_{5,4,1}^{C,out}$	0.0
$b_{1,5,1}^{C,out}$	0.0	$b_{5,4,2}^{C,out}$	0.0
$b_{1,5,2}^{C,out}$	0.0	$b_{5,5,1}^{C,out}$	0.0
$b_{2,1,1}^{C,out}$	0.0		
$b_{2,1,2}^{C,out}$	0.0		
$b_{2,2,2}^{C,out}$	705.1600000000011		
$b_{2,3,2}^{C,out}$	0.0		
$b_{2,4,1}^{C,out}$	0.0		
$b_{2,4,2}^{C,out}$	0.0		
$b_{2,5,1}^{C,out}$	785.3669846984699		